Self-Supervised Learning With Adaptive Distillation for Hyperspectral Image Classification

Jun Yue, Leyuan Fang, *Senior Member, IEEE*, Hossein Rahmani, and Pedram Ghamisi, *Senior Member, IEEE*

***Abstract*— Hyperspectral image (HSI) classification is an important topic in the community of remote sensing, which has a wide range of applications in geoscience. Recently, deep learning-based methods have been widely used in HSI classification. However, due to the scarcity of labeled samples in HSI, the potential of deep learning-based methods has not been fully exploited. To solve this problem, a self-supervised learning (SSL) method with adaptive distillation is proposed to train the deep neural network with extensive unlabeled samples. The proposed method consists of two modules: adaptive knowledge distillation with spatial–spectral similarity and 3-D transformation on HSI cubes. The SSL with adaptive knowledge distillation uses the self-supervised information to train the network by knowledge distillation, where self-supervised knowledge is the adaptive soft label generated by spatial–spectral similarity measurement. The SSL with adaptive knowledge distillation mainly includes the following three steps. First, the similarity between unlabeled samples and object classes in HSI is generated based on the spatial–spectral joint distance (SSJD) between unlabeled samples and labeled samples. Second, the adaptive soft label of each unlabeled sample is generated to measure the probability that the unlabeled sample belongs to each object class. Third, a progressive convolutional network (PCN) is trained by minimizing the cross-entropy between the adaptive soft labels and the probabilities generated by the forward propagation of the PCN. The SSL with 3-D transformation rotates the HSI cube in both the spectral domain and the spatial domain to fully exploit the labeled samples. Experiments on three public HSI data sets have demonstrated that the proposed method can achieve better performance than existing state-of-the-art methods.**

***Index Terms*— Deep neural network, hyperspectral image (HSI) classification, knowledge distillation, self-supervised learning (SSL), spatial–spectral feature extraction.**

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I. Introduction

HYPERSPECTRAL images (HSIs) usually contain hundreds of spectral channels, and each pixel in HSI has a corresponding spectral curve [1], [2]. Combining both spatial texture information and spectral reflectance information, HSIs have been widely used in many fields, e.g., agriculture, marine monitoring, geological exploration, environment, and ecology. HSI classification is utilized to determine the corresponding object category for each pixel (e.g., soil, trees, grass, building, roads, and river) which plays a key role in many remote sensing applications [3], [4].

In recent decades, many HSI classification methods have been proposed, including spectral matching-based methods, which determine the object types of samples based on the matching degree of spectral curves. Typical spectral matching-based methods include spectral angle mapper (SAM) [5], [6] and spectral information divergence (SID) [7]. Due to the high spectral dimensionality in HSIs, many statistical feature reduction methods have been proposed to transform spectral vector of HSI from high-dimensional feature space to low-dimensional feature space, including linear discriminant analysis [8], principal components analysis (PCA) [9], independent component analysis (ICA) [10], and minimum noise fraction (MNF) [11]. In addition, because of the nonlinear characteristics of HSI, many nonlinear feature extraction methods have been proposed in recent years, such as manifold coordinate representations [12]–[14], locality-preserving discriminant analysis [15], locality preserving projections (LPP) [16], and sparsity preserving projections (SPP) [17].

Due to the characteristics of spatial homogeneity and heterogeneity of HSI, it is hard to make full use of the features of HSI only by spectral feature extraction. Therefore, researchers have proposed a series of joint spatial–spectral feature extraction methods [18], [19]. The typical one is extended morphological profile (EMP) [20]. The first step of this method is to extract features by PCA and then generate spatial features by morphological operations (opening or closing operations). Recently, morphological-based joint spatial–spectral feature extraction methods have been continuously improved, including extended attribute profile (EAP) [21], and directional morphological profiles (DMP) [22].

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With the introduction and rapid development of deep learning [23]–[25], it has achieved great success in many tasks such as image classification [26]–[28], object detection [29]–[31], image segmentation [32]–[34], scene recognition [35], natural language processing [36], [37], optical character recognition [38], and so on. At the same time, researchers have proposed a series of deep learning-based methods for HSI classification, and the classification accuracy of HSI has been gradually improved [18]. To fully exploit spatial–spectral features of HSI, some joint spatial–spectral feature extraction methods with a deep neural network have been proposed [39], including stacked autoencoders [40], deep belief network [41], superpixel-based discriminative sparse model [42], and local covariance matrix representation [43].

However, the application of deep learning for HSI classification has not been well-addressed. One of the important reasons is that the deep learning-based method requires a large number of labeled samples to obtain satisfactory accuracy. In the process of supervised HSI classification, there are usually two ways to obtain training samples: 1) field investigation and 2) visual interpretation directly from high-resolution images. In particular, the training samples collected through field investigation can usually lead to higher classification accuracy [44]. However, compared with visual interpretation, field investigation is costly, complex, and time-consuming, which greatly limits the number of training samples. In fact, it is difficult to obtain enough training samples to fully meet the training requirements of the deep neural network, which limits the researchers' attempts to further improve the classification accuracy of HSI using a deep neural network.

To solve this problem, several HSI classification methods based on few-shot learning have been proposed. Few-shot learning is an important branch of machine learning which is designed to address the problem of limited training samples. Usually, metric-learning and meta-learning strategies are used to learn the features of HSIs, and the absolute distance or relative distance between features is used to identify the object class of each unlabeled pixel in HSI [45], [46]. However, these methods still do not solve the problem of lack of samples very well. As a result, researchers began to explore some solutions to get supervision from the data itself, one of which is self-supervised learning (SSL). SSL mines its own supervised information from large-scale unsupervised data. In other words, the supervised information used in SSL does not need to be obtained by manual annotation, but the algorithm automatically constructs the supervised information from unsupervised data to train the model [47], [48]. The existing SSL methods include the method based on spatial relationship [49], [50], inpainting [51], image reconstruction [52], color transformation [53], super resolution [54], and spatial rotation transformation [55], [56]. In the field of HSI classification, some super-pixel-based SSL methods have been proposed to fully exploit each labeled sample [57], [58]. However, none of the above methods generate corresponding soft labels for unlabeled samples and use them to train the network. From the perspective of training data transformation, the existing SSL-based methods for image classification are based on transformation in 2-D space. Since hyperspectral data is a 3-D cube with 2-D spatial dimension and spectral dimension, it is necessary to update the SSL with 2-D transformation to an SSL with 3-D transformation.

In this article, we propose a self-supervised learning method with adaptive distillation (SSAD) to train a deep neural network with a large number of unlabeled samples. The SSAD mainly consists of two modules: 1) SSL with adaptive knowledge distillation and 2) SSL with 3-D transformation. In the adaptive knowledge distillation, the unlabeled samples are adaptively labeled based on the spatial–spectral distance measurement. By calculating the similarity between unlabeled samples and predefined object classes, the adaptive soft label of each unlabeled sample is generated. The adaptive soft label includes the probability that the sample belongs to each predefined object class. The proposed adaptive knowledge distillation method mainly includes the following steps. First, in order to measure the similarity between samples in the HSI, the spatial–spectral joint distance (SSJD) considering both spectral Kullback–Leibler (KL) divergence and spatial Euclidean distance is calculated. Second, based on the SSJD between unlabeled samples and labeled samples, the distance between unlabeled samples and pre-defined object classes in HSI is generated. Third, the adaptive soft label of each unlabeled sample is generated by the distance-probability transformation method. Fourth, the proposed progressive convolutional network (PCN) is trained by minimizing the cross-entropy between the probabilities generated by PCN and the adaptive soft label of each unlabeled sample. In addition, SSL with 3-D transformation further exploits each labeled sample by transforming HSI in spectral and spatial domains.

The main contributions of this article are threefold as follows:

1) An unsupervised similarity measurement method between samples is proposed. This method considers both the spectral distance and the spatial distance between samples in HSI. Based on the SSJD, the similarity between each unlabeled sample and each predefined object class is generated.
2) An SSL strategy with an adaptive soft label is proposed. The adaptive soft label of each unlabeled sample measures the probability that each unlabeled sample belongs to each predefined object class. We add the adaptive soft label to the FCN model training process by calculating the cross-entropy between the adaptive soft label and the output of the FCN model.
3) An SSL strategy with 3-D transformation is proposed by rotating the original HSI image in spectral and spatial domains.

This article is organized as follows. In Section II, the proposed method is explained in detail, including SSJD, adaptive soft label and adaptive knowledge distillation, and the SSL strategy with 3-D transformation. In Section III, the experimental data and parameters are described. In Section IV, the results are shown and discussed, and the ablation study is analyzed. In Section V, the conclusions of this article are summarized.

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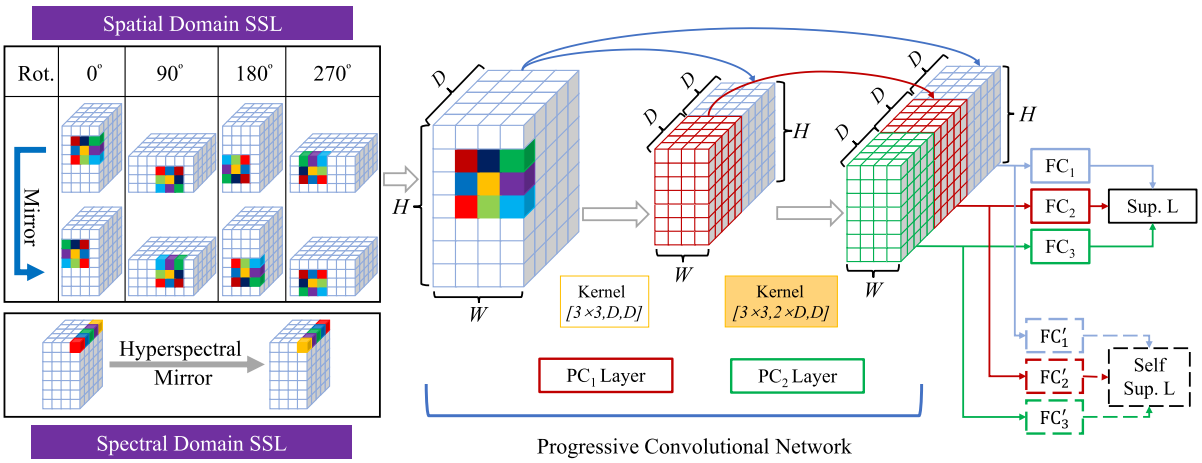


Fig. 1. Self-supervised learning with 3-D transformation and progressive convolutional network, where PC and FC are abbreviations for progressive convolution and fully connected layer, respectively, and D is the number of spectral channels of the input HSI.

II. Method

In this article, an SSL method is proposed, which consists of two main modules: SSL with 3-D transformation and SSL with adaptive knowledge distillation. These two modules improve the existing SSL methods from the aspects of data input, self-supervised label generation, and self-supervised training.

A. SSL With 3-D Transformation

The existing SSL methods include the approaches that are based on spatial relationship [49], [50], the approaches that are based on spatial rotation transformation [55], [56], etc. However, these methods are based on operations in 2-D space. This article presents an SSL method for geometric transformation in 3-D space. Since HSIs are 3-D cubes with 2-D spatial and 1-D spectral dimensions, they can be rotated and flipped in 3-D space. The details of the method are shown in Fig. 1.

1) Spatial and Spectral Domain Geometric Transformation: Different from the nature images, the HSIs are rotational invariance and symmetry in the spatial domain. In other words, the horizontal rotation in the spatial domain does not change the predictions of the pixels in HSIs. Therefore, we rotate and mirror the HSI in the spatial domain to promote the robustness of the HSI classification model. The cross-entropy loss can be calculated by rotating the ground-truth map with the same horizontal rotation operation as the input HSI. In our implementation, we define four horizontal rotation operations, and the corresponding rotation angle set is $\mathcal{T} = \{90° \cdot i \mid i \in [0, 1, 2, 3]\}$. Specifically, given the HSI $\mathcal{H}$, we first generate its four horizontal rotated replicas $\{\mathcal{H}^t \mid t \in \mathcal{T}\}$, where $\mathcal{H}^t$ is the replica of HSI $\mathcal{H}$ rotated by t degrees. We define mirroring in a 2-D spatial dimension as a horizontal mirror. For each HSI $\mathcal{H}^t$, we perform a horizontal mirror operation as shown in Fig. 1 to generate the corresponding mirroring HSI $\bar{\mathcal{H}}^t$.

Since HSIs have three dimensions (i.e., 2-D spatial dimension and 1-D spectral dimension), this article proposes to rotate HSIs in the spectral domain and predict the order of spectral sequence. We define two kinds of spectral sequences. One is frequency from high to low, and its label is defined as 1. The other is frequency from low to high, and its label is defined as 0. We define mirroring in spectral dimension as hyperspectral mirror operation. Combining three operations (i.e., horizontal rotation, horizontal mirror, and hyperspectral mirror), the training set can be denoted as $\{\mathcal{H}_1^t \cup \bar{\mathcal{H}}_1^t \cup \mathcal{H}_0^t \cup \bar{\mathcal{H}}_0^t \mid t \in \mathcal{T}\}$, where $\mathcal{H}_1^t$ is the HSI $\mathcal{H}$ rotated by t degrees, and its spectral sequence is arranged from high frequency to low frequency. $\mathcal{H}_0^t$ is the HSI $\mathcal{H}$ rotated by t degrees, and its spectral sequence is arranged from low frequency to high frequency. $\bar{\mathcal{H}}_1^t$ is the HSI $\bar{\mathcal{H}}$ rotated by t degrees, and its spectral sequence is arranged from high frequency to low frequency. $\bar{\mathcal{H}}_0^t$ is the HSI $\bar{\mathcal{H}}$ rotated by t degrees, and its spectral sequence is arranged from low frequency to high frequency.

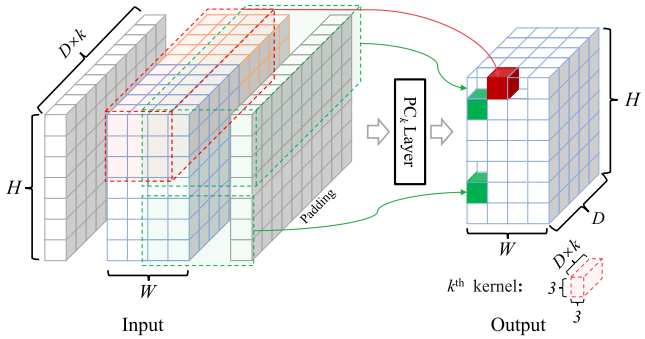


Fig. 2. Operations of the kth progressive convolution layer. The size of the input matrix and the output matrix of the kth PC layer are $[W, H, k \times D]$ and $[W, H, D]$, respectively. The input matrix of the kth PC layer is the concatenation of the output matrices of previous layers.

2) Progressive Convolutional Network: In order to realize the geometric transformation in spatial and spectral domains, a fully convolutional neural network called PCN is designed. In the PCN, the output of the forward propagation rotates with the HSI. The operations of the kth progressive convolution (PC) layer are shown in Fig. 2. In the kth PC layer, the input matrix size is $W \times H$ and the number of channels is $D \times k$, where W and H are the width and height of the HSI,

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respectively. The input matrix of the kth PC layer is formed by concatenating the outputs of the 0th to $(k-1)^{\text{th}}$ PC layer. First, we use the padding strategy around the input of the 0th PC layer. Second, for the $(k-1)$th PC layer, we employ D PC filters to generate the feature matrix $\mathcal{O}_{k-1}$. Third, for the kth PC layer, we concatenate the outputs across all the previous PC layers into a matrix with a size of $W \times H \times D \times k$ as the input of the kth PC layer. The operations of the 0th PC layer of PCN can be formulated as follows:

$$\begin{cases} \mathcal{I}_0 = \mathcal{H}, & k = 0 \\ \mathcal{O}_0 = \mathrm{PC}_0(\mathcal{I}_0), & k = 0 \end{cases} \tag{1}$$

where $\mathcal{O}_0$ represents the output of the 0th PC layer. Then, the input and output of the kth PC layer can be formulated as follows:

$$\begin{cases} \mathcal{I}_k = \left[\mathcal{O}_0, \mathcal{O}_1, \ldots, \mathcal{O}_{k-1}\right], & k > 0 \\ \mathcal{O}_k = \mathrm{PC}_k(\mathcal{I}_k), & k > 0 \end{cases} \tag{2}$$

where $\mathcal{I}_k$ and $\mathcal{O}_k$ is the input and output of the kth PC layer, respectively. In each PC layer, the activation function ReLU (rectified linear unit) [59] is applied to promote the learning ability of the PCN. In addition, the dropout strategy [25] is adopted to improve the generalization ability of the model and avoid over-fitting. In convolutional neural networks, the deeper the layers, the larger the receptive field. Therefore, the proposed PCN structure can perform convolution operation on multiscale receptive fields and extract multiscale features effectively.

3) Multi-PC Layers Fusion: The SSL with 3-D transformation strategy enlarges the input hyperspectral data, and the corresponding classification results are expanded accordingly. In each PC layer, we introduce a fully connected (FC) layer to transfer each pixel feature into class predictions (logits). The calculations are defined as follows:

$$\alpha_k = \mathrm{FC}_k(\mathcal{O}_k) = \mathcal{O}_k \cdot W_k + b_k, \quad k > 0 \tag{3}$$

where α_k is the prediction of the kth PC layer and W_k and b_k are the parameters of the kth FC layer. Thus, given the inputs $\{\mathcal{H}_1^t \cup \tilde{\mathcal{H}}_1^t \cup \mathcal{H}_0^t \cup \tilde{\mathcal{H}}_0^t \mid t \in \mathcal{T}\}$ of PCN, the output α_k of the kth PC layer is a set of logits, i.e., $\mathcal{Q}_k = \{\alpha_k^t \cup \bar{\alpha}_k^t \mid t \in \mathcal{T}\}$. We then fuse the set of logits to generate the merged result of the kth PC layer by

$$\tilde{\alpha}_k = \frac{1}{|\mathcal{Q}_k|} \sum \$\left(\dot{\alpha}_k^t\right) \quad \forall \dot{\alpha}_k^t \in \mathcal{Q}_k \tag{4}$$

where $\$$ is the *softmax* operation and $|\mathcal{Q}_k|$ is the size of the logits set.

Furthermore, we fuse the output logits generated from multi-PC layers. Given the output logits set $\mathcal{R} = \{\tilde{\alpha}_k \mid k \in [1, 2, \ldots N_k]\}$ which were generated by the 1st to the N_kth PC layers, we use the *softmax* function to normalize each logits vector and sum all normalized logits vectors from multilayers as follows:

$$\alpha_o = \frac{1}{N_k} \sum_{k=1}^{N_k} \$(\tilde{\alpha}_k) \quad \forall \tilde{\alpha}_k \in \mathcal{R} \tag{5}$$

where α_o is the final prediction of our proposed network and N_k is the total number of PC layers. Finally, we apply the *argmax* function on α_o and generate the HSI pixel-level labels with the maximum logit value.

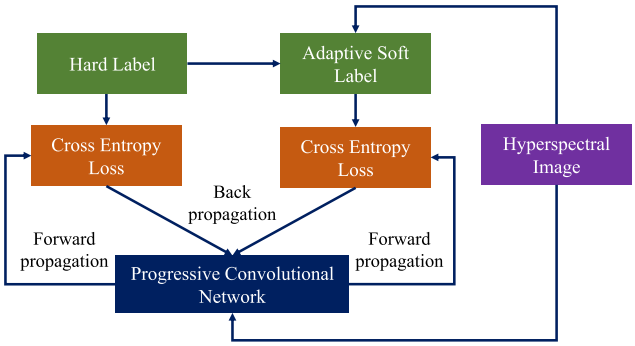


Fig. 3. Pipeline of SSL with adaptive knowledge distillation.

B. SSL With Adaptive Knowledge Distillation

Knowledge distillation is a training strategy by transferring the knowledge from one teacher network with higher precision and larger model size to a student network with smaller model size. Knowledge distillation uses a larger network to generate soft label, which is used to guide the training of the student network. In this article, adaptive knowledge distillation is proposed by transferring the self-supervised knowledge to the network with adaptive soft and hard labels. The flowchart of SSAD distillation is shown in Fig. 3. The SSAD method consists of two steps. The first step is to generate adaptive soft labels for all unlabeled samples. The second step is to distill the self-supervised adaptive soft labels into the network.

Algorithm 1 Adaptive Soft Label Generation

Require:
The labels set $\mathcal{C}$
The labeled samples set $\mathcal{L}$
The unlabeled samples set $\mathcal{U}$
Ensure:
The adaptive soft labels set $\mathcal{S}$;
for $u_i \in \mathcal{U}$ **do**
  for $l_i \in \mathcal{L}$ **do**
    Compute the Euclidean Distance between u_i and l_i
    Compute the Kullback-Leibler divergence between u_i and l_i
    Compute SSJD(u_i, l_i) between u_i and l_i by (6)
  end for
end for
for $u_i \in \mathcal{U}$ **do**
  for $c_i \in \mathcal{C}$ **do**
    Compute the average SSJD(u_i, c_i) between u_i and class c_i by (8)
    Calculate the probability $P(\Phi(u_i) = c_i)$ that u_i belongs to class c_i by (9)
  end for
  Generate the adaptive soft label $\mathcal{S}(u_i)$ by (10)
  Add $\mathcal{S}(u_i)$ to adaptive soft label set $\mathcal{S}$
end for

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1) Adaptive Soft Label Generation: Given the HSI $\mathcal{H}$, we denote its labeled samples set, unlabeled samples set, and labels set as $\mathcal{L}$, $\mathcal{U}$, and $\mathcal{C}$, respectively. In order to label all unlabeled samples based on spectral–spatial correlation between $\mathcal{L}$ and $\mathcal{U}$, we first define the SSJD between samples. SSJD is composed of spatial distance and spectral distance, which can be calculated by

$$\mathrm{SSJD}(u_i, l_i) = \sqrt{\mathrm{ED}(u_i, l_i) \cdot \mathrm{SID}(u_i, l_i)} \tag{6}$$

where $l_i \in \mathcal{L}$, $u_i \in \mathcal{U}$, $\mathrm{ED}(u_i, l_i)$ is the Euclidean distance between l_i and u_i, and $\mathrm{SID}(u_i, l_i)$ is the SID [10] between u_i and l_i, which can be formulated as:

$$\mathrm{SID}(u_i, l_i) = \mathrm{KL}(u_i \parallel l_i) + \mathrm{KL}(l_i \parallel u_i) \tag{7}$$

where $\mathrm{KL}(l_i \parallel u_i))$ denotes the relative entropy of u_i with respect to l_i, which is also known as Kullback–Leibler divergence. Based on SSJD between samples, we define the SSJD between sample and class, which can be formulated as:

$$\mathrm{SSJD}(u_i, c_i) = \sum_{r=1}^{N_{c_i}} \frac{\mathrm{SSJD}(u_i, l_r)}{|\mathcal{C}|^{\mathrm{n}(u_i, l_r)}} \quad \forall \Phi(l_r) = c_i \tag{8}$$

where $c_i \in \mathcal{C}$, $\Phi(l_r)$ denotes the label of l_r, $|\mathcal{C}|$ is the size of the labels set, N_{c_i} is the total number of labeled samples which belongs to class c_i, and $\mathrm{n}(u_i, l_r)$ is the ordinal by sorting all SSJDs between u_i and $l_r, \forall \Phi(l_r) = c_i$. Given the distance between sample and class, the probability that unlabeled sample u_i belonging to class c_i can be formulated as:

$$\mathrm{P}(\Phi(u_i) = c_i) = \frac{e^{-\mathrm{SSJD}(u_i, c_i)|\mathcal{C}|}}{\sum_{r=1}^{|\mathcal{C}|} e^{-\mathrm{SSJD}(u_i, c_r)|\mathcal{C}|}} \tag{9}$$

where e is the natural constant. Finally, the adaptive soft label of u_i can be generated by

$$\mathcal{S}(u_i) = [\mathrm{P}(\Phi(u_i) = c_1), \mathrm{P}(\Phi(u_i) = c_2), \ldots, \mathrm{P}\big(\Phi(u_i) = c_{|\mathcal{C}|}\big)] \tag{10}$$

The procedure of the adaptive soft label generation is given in Algorithm 1.

2) Adaptive Knowledge Distillation: We design an optimization method called adaptive knowledge distillation to help the PCN learn from both the hard labels and the adaptive soft labels. We first calculate the loss between the prediction α_k and the hard labels $\mathcal{G}$:

$$\mathcal{L}_k^{\mathcal{G}} = \Delta(\alpha_k, \mathcal{G}) = \Delta(\delta(\alpha_k), \mathcal{G}) \tag{11}$$

where Δ is *cross-entropy* function and δ is *softmax* operation.

Second, we calculate the loss between the prediction α_k and the adaptive soft labels $\mathcal{S}$ by

$$\mathcal{L}_k^{\mathcal{S}} = \Delta(\delta(\alpha_k), \mathcal{S}) \tag{12}$$

In addition, we introduce another FC layer after each PC layer to transfer each pixel feature into spectral sequence predictions. The forward propagation of spectral sequence prediction is formulated by

$$s_k^s = \mathrm{FC}_k^s(O_k) = O_k \cdot W_k^s + b_k^s, \quad k > 0 \tag{13}$$

TABLE I

DETAILS OF THE HSI DATA SETS, WHERE W, H, AND D ARE THE WIDTH, HEIGHT, AND NUMBER OF SPECTRAL CHANNELS OF THE HSI, RESPECTIVELY. N_C IS THE NUMBER OF OBJECT CATEGORIES, AND N_A IS THE NUMBER OF ACTIVATING PIXELS

Datasets	W	H	D	N_C	N_A
IP	145	145	200	16	10249
UP	610	340	103	9	42776
HS	1905	349	144	15	15029

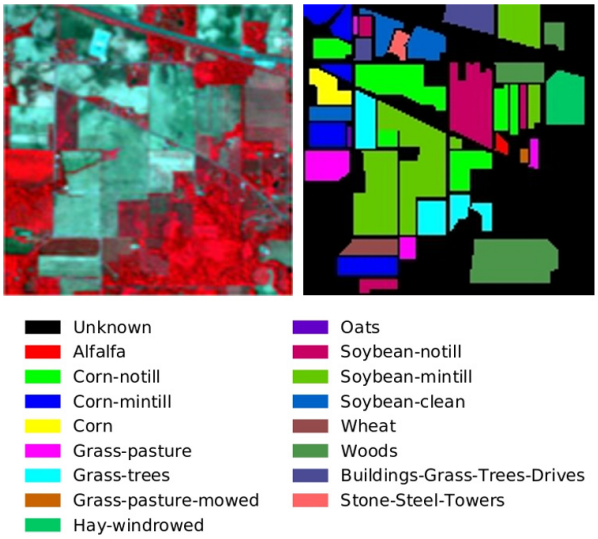


Fig. 4. False color composite image and the corresponding ground-truth map of the IP data set.

where W_k^s and b_k^s are the weights and bias of the kth FC layer for predicting the spectral sequence, respectively. The self-supervised loss on spectral domain can be formulated as follows:

$$\mathcal{L}_k^S = \Delta\big(\alpha_k^s, y\big) \tag{14}$$

where $y \in \{0, 1\}$ represents the order of the spectral sequence. $y = 1$ indicates that the frequency of the input hyperspectral data is from high to low. Otherwise, $y = 0$ indicates that the frequency of the input hyperspectral data is from low to high. Finally, the total loss for training PCN is defined as follows:

$$\mathcal{L} = \frac{1}{N_k} \sum_{k=1}^{N_k} \big(\mathcal{L}_k^G + \mathcal{L}_k^P + \mathcal{L}_k^S\big) \tag{15}$$

where N_K is the total number of PC layers.

III. EXPERIMENTS

A. Data Sets and Evaluation

To demonstrate the effectiveness of the proposed method, experiments are conducted on three well-known HSI data sets including Indian Pines (IP), University of Pavia (UP), and Houston (HS). The details of the three data sets are listed in Table I.

1) IP data set: This data set was collected by Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensors

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Fig. 5. False color composite image and the corresponding ground-truth map of the UP data set.

in Northwestern Indiana. The spatial resolution of this image is 20 m covering spectral channels from 200 to 2400 nm. This image is consisting of 145 by 145 pixels. Before data preprocessing, the spectral channels affected by noise and water absorption (104–108, 150–163, and 220) are removed, and the remaining 200 spectral channels are involved in the experiment. The corresponding ground-truth of the IP data set contains 16 object categories. The false color composite image of the IP data set and its corresponding ground-truth map are shown in Fig. 4.

2) UP data set: This data set was collected by the Reflective Optics Systems Imaging Spectrometer (ROSIS) sensor in Pavia, Italy. This image is mainly composed of urban features around the University of Pavia with 610 by 340 pixels. The spatial resolution of this image is 1.3 m covering spectral bands from 430 to 860 nm. The UP data set contains 115 spectral bands, 12 of which are removed because of noise and water absorption, and the remaining 103 spectral channels are involved in the experiment. The corresponding ground-truth of the UP data set contains nine object categories. The false color composite image and the corresponding ground-truth map of the UP data set are shown in Fig. 5.
3) HS data set: This data set was collected by the ITRES-CASI 1500 sensor [60]. This image is mainly composed of urban features around the University of Houston with 349 by 1095 pixels. The spatial resolution of this image is 2.5 m covering spectral bands from 380 to 1050 nm. HS data set contains 144 spectral bands. The corresponding ground-truth of the UP data set contains 15 object categories. The false color image and the corresponding ground-truth map of the HS data set are shown in Fig. 6.

The proposed method was evaluated on several tasks. In each task, L labeled samples of each object category are randomly sampled for training. From the activating samples other than the training samples, three labeled samples of each object category are randomly sampled for validation, and the rest of the activating samples are used for testing. To report the classification results, we conduct ten independent runs for each task and compute the mean OA (overall accuracy), AA (average accuracy), and kappa coefficient with standard deviation over the ten runs.

B. Parameter Settings

Defining the format of the parameters of the kth($k > 0$) PC layer as $[kernel\ size, N_i, N_o, padding, stride]$, where N_i and N_o are the number of input filters and output filters, respectively. The proposed PCN is a fully convolutional neural network, and the input size of the PCN is $W \times H \times D$ for each epoch. Specifically, the input sizes of IP, UP, and HS data sets are $145 \times 145 \times 200$, $610 \times 340 \times 103$, and $1905 \times 349 \times 144$, respectively. We set the parameters of the kth PC layer as $[3 \times 3, k \times D, D, 1, 1]$. As a result, the size of the input matrix and the output matrix of the kth PC layer are $[W, H, k \times D]$ and $[W, H, D]$, respectively. In our proposed PCN, the embedding operation of each PC layer is transferring a matrix with N_i filters into a matrix with N_o filters. Thus, given the HSI with D-bands and N_C categories, the parameters W and b of each FC are a matrix in $\mathbb{R}^{D \times N_C}$ and a vector in $\mathbb{R}^{N_C}$, respectively.

We set the range of the depth N_K of our proposed network from 1 to 15 and train these PCNs only under the cross-entropy loss by (11). While training the model, we remove the SSL with a 3-D transformation module. For each depth of PCN, we conduct ten independent runs, and the average accuracies over ten runs as well as the 70% confidence intervals are shown in Fig. 7. For each run, we fuse the output predictions of all PC layers by (5). As shown in Fig. 7, with the increase of the depth of PCN, the average accuracy of the PCN rises first and falls later. As the depth of PCN increases, the receptive field increases and the oversized receptive field results in the redundancy of input information. Therefore, we set the depth of PCN to be 3 since the PCN with three PC layers requires less computation and achieves better performance.

In the proposed PCN, the parameter of the dropout (dropout probability) is $\rho = 0.5$, and all the dropout layers are removed in the testing stage. To avoid outliers, for each unlabeled sample, we set the minimum SSJD to be no larger than 0.085. During the training stage, we train our network by (15) and use Adam to perform stochastic optimization. The training learning curve and the validation learning curve are shown in Fig. 8 under a learning rate of 10^{-4}. As shown in Fig. 8, the learning curves show that when the learning rate is 10^{-4}, a stable training loss and validation loss can be obtained without overfitting. We set the initial learning rate and the weight decay to 10^{-4} and 10^{-5}, respectively. After every 1000 iterations, the learning rate is reduced by 0.1.

IV. Results and Discussion

A. Accuracies

In this study, for the IP and UP data sets, the experiments were conducted with the L number of supervised samples for classification as 5 and 10, respectively, due to the limited number of labeled samples for a class (i.e., "Oats"). For the HS data set, L was set to 50 and 80, respectively, because this data set had a relatively larger number of training samples. The classification maps are shown in Figs. 9–11.

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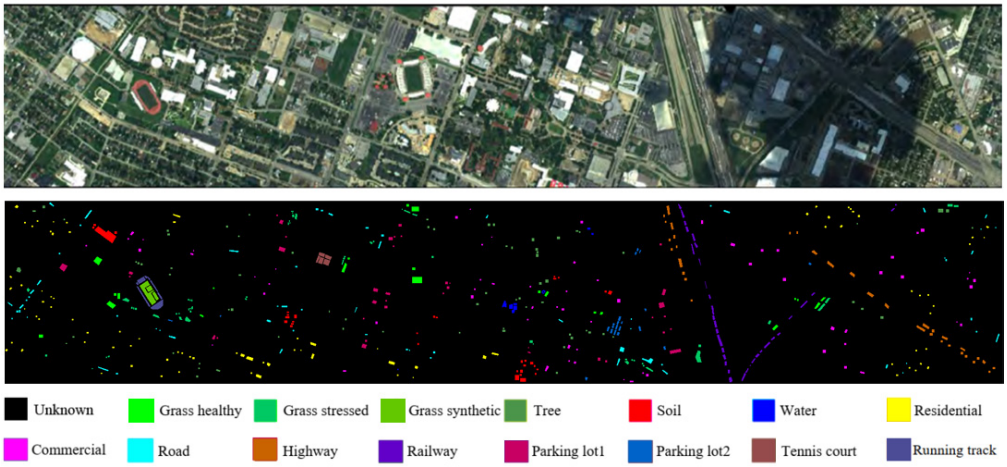


Fig. 6. False color composite image and the corresponding ground-truth map of the HS data set.

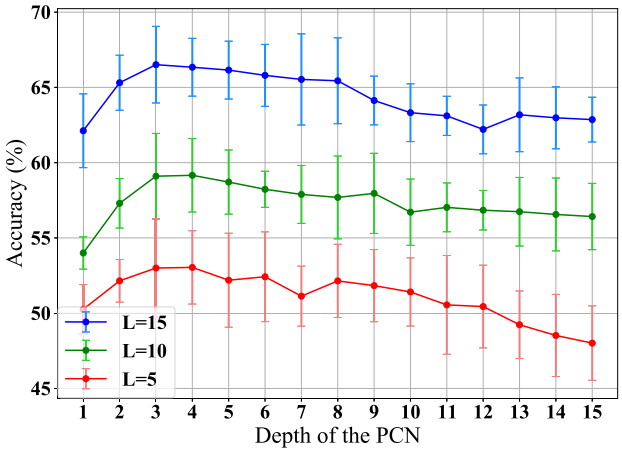


Fig. 7. Accuracies of PCN with different layers on IP data set.

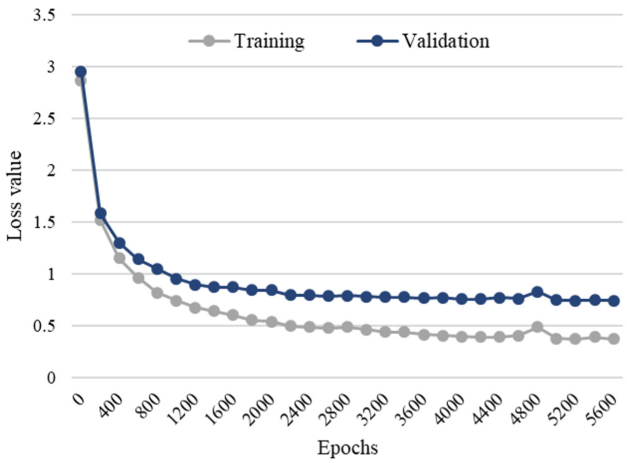


Fig. 8. Learning curves under a learning rate of 10^{-4}. The loss values are recorded every 200 epochs.

We compare our method with state-of-the-art HSI classification ones under the small-scale training samples. The compared methods include 2D-CNN, 3D-CNN [61], DRNN (deep recurrent neural networks) [60], DBMA (double-branch multiattention mechanism network) [62], MSDN (end-to-end 3-D dense convolutional network) [63], and DFSL (deep few-shot learning for HSI classification) [45]. The comparisons of IP, UP, and HS data sets are shown in Tables II–IV, respectively.

B. Ablation Study

The proposed method in this article mainly consists of two strategies, i.e., SSL with 3-D transformation and adaptive knowledge distillation. To demonstrate the effectiveness of each strategy, an ablation study was performed on IP data set. In other words, the accuracy of the proposed method was tested when the two strategies were removed.

1) Effectiveness of SSL With 3-D Transformation: In this ablation study, the SSL with 3-D transformation is used in both the training and the testing stages. Thus, the output of PCN with three PC layers is a set of logits: $\mathcal{Q} = \{\alpha_k^t \cup \bar{\alpha}_k^t \mid t \in \mathcal{T}, k \in [1, 2, 3]\}$. Then, the output logits are fused by (4) to generate the prediction $\tilde{\alpha}_k$ of the kth PC layer. The accuracies are shown in Table V, where "w/ SSL-3DT" means the network was trained with the 3-D transformation strategy (3DT). The "baseline" method is a pure three-layer PCN without the 3-D transformation strategy. Compared with the results generated by the "baseline" method, the results generated by PCN with the 3-D transformation strategy perform better in all settings. Meanwhile, the results generated by the deeper PC layer are higher than those generated by previous PC layers. Finally, we fuse the output logits of all the PC layers by (5) to generate the fused results: α_o. As shown in Table V, the fused results α_o perform the best accuracies. These improvements indicate the usefulness of SSL with 3-D transformation.

2) Effectiveness of SSL With Adaptive Knowledge Distillation: In this ablation study, we train the PCN with adaptive knowledge distillation but without 3-D transformation. The classification results are shown in Table VI, where "w/ SSL-AKD" means the PCN trained under the strategy of SSL with the adaptive knowledge distillation (AKD). The "baseline" method is a pure three-layer PCN without the adaptive

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TABLE II

CLASSIFICATION RESULTS (%) USING FIVE AND TEN LABELED SAMPLES FOR EACH CLASS ON THE IP DATA SET (THE BOLD VALUE IS THE BEST ACCURACY IN EACH CASE)

Class	2D-CNN		3D-CNN		DRNN		DBMA		MSDN		DFSL		SSAD(Ours)	
	L=5	L=10	L=5	L=10	L=5	L=10	L=5	L=10	L=5	L=10	L=5	L=10	L=5	L=10
1	73.3±3.1	77.3±2.7	80.8±2.9	88.9±2.7	79.5±3.9	87.0±3.5	91.8±3.2	92.5±4.4	85.0±4.7	95.7±1.3	83.3±1.6	92.7±1.8	97.3±1.8	99.3±0.3
2	62.2±9.6	71.0±8.7	61.3±8.7	69.4±9.2	71.0±7.6	79.3±5.3	59.7±11.9	67.3±10.6	54.8±12.0	70.5±5.6	53.1±7.1	64.1±5.9	70.4±4.1	80.9±2.5
3	55.1±9.2	61.0±8.2	67.4±6.7	65.3±9.1	62.2±8.6	67.3±7.6	64.9±10.9	73.5±9.6	61.4±10.9	74.7±4.8	63.7±7.3	70.7±5.4	75.1±3.8	79.7±2.3
4	63.0±5.2	65.7±7.6	69.0±8.1	73.5±1.6	67.6±3.7	74.9±4.9	89.3±2.1	79.8±2.6	81.1±7.3	85.3±4.7	90.1±5.6	90.7±3.9	79.8±6.7	83.5±3.4
5	60.6±9.1	63.6±9.1	70.4±7.9	68.7±8.2	65.2±8.7	71.9±8.4	66.8±8.3	74.8±12.7	60.2±7.2	75.3±8.0	65.0±10.0	71.0±5.4	75.0±1.9	82.1±2.3
6	62.4±7.1	66.0±9.3	72.4±6.4	70.6±9.5	66.4±7.5	73.7±6.6	66.5±13.1	78.7±11.8	65.1±15.6	80.0±6.6	71.7±8.3	69.3±7.9	81.6±6.2	83.0±2.4
7	70.9±4.5	70.2±4.4	82.8±5.1	75.1±5.1	74.5±4.8	81.9±4.9	91.3±1.3	73.1±2.5	83.9±2.5	84.8±0.0	90.7±3.1	100.0±0.0	92.6±3.0	89.4±3.6
8	66.0±6.1	78.0±5.3	80.7±9.5	86.8±4.0	78.0±4.4	84.1±5.0	90.4±3.7	90.7±3.8	84.2±6.0	92.4±2.8	91.9±3.5	90.5±3.4	95.3±4.5	97.8±2.5
9	80.3±4.6	89.6±2.5	83.6±2.3	100.0±0.0	89.3±3.3	98.5±1.2	85.9±7.1	84.6±6.9	81.1±13.9	90.8±1.4	94.7±1.8	89.5±4.9	95.1±7.2	96.0±1.4
10	53.0±9.4	60.8±9.2	57.5±8.3	67.6±11.0	59.8±8.1	65.3±8.9	69.2±15.6	74.6±13.7	70.7±15.5	77.7±6.7	68.6±8.5	75.4±7.4	76.3±4.2	80.4±2.4
11	63.5±8.4	68.7±8.7	59.2±7.5	68.8±7.7	71.0±8.4	73.3±7.7	61.8±9.9	75.1±16.0	63.4±14.2	75.2±8.0	60.9±10.2	71.2±7.4	81.1±2.5	82.9±1.9
12	52.5±7.7	68.7±6.1	52.9±9.5	67.2±4.3	65.0±6.6	72.3±7.2	71.9±5.2	74.8±4.5	64.2±8.3	77.2±3.4	66.2±4.2	77.0±2.5	75.2±2.6	81.2±2.1
13	66.7±7.5	72.1±4.3	75.5±1.4	77.3±2.5	73.0±4.2	71.6±4.2	88.9±4.4	82.8±4.9	86.8±2.9	88.5±1.2	85.2±1.6	92.4±1.7	93.9±1.6	91.5±2.6
14	73.7±4.8	79.7±4.6	81.6±5.9	78.8±3.1	80.6±3.9	88.3±2.6	85.5±3.1	85.1±2.6	79.8±4.7	88.8±1.7	86.2±2.2	86.6±2.1	88.6±3.1	91.8±2.1
15	66.9±6.2	70.2±2.5	79.4±3.1	73.5±2.1	71.7±3.5	79.8±3.6	89.3±2.4	80.2±3.5	83.9±5.3	87.5±1.1	94.8±1.4	90.9±2.7	88.4±3.5	87.1±2.3
16	67.0±8.7	70.9±7.1	91.0±2.2	83.6±1.2	72.0±3.5	80.5±2.2	90.7±3.1	94.8±3.1	85.4±6.0	99.0±0.4	84.9±1.1	82.5±2.7	89.8±2.4	97.7±2.0
OA	62.4±5.5	69.1±4.8	66.6±4.0	71.3±3.5	69.8±5.8	75.5±4.3	70.5±5.9	76.7±5.4	67.7±6.3	78.9±2.4	69.4±3.1	75.4±2.9	**80.3±2.5**	**84.3±2.0**
AA	64.8±4.9	70.8±4.6	72.8±3.7	75.9±2.5	71.7±4.4	78.1±3.4	79.0±4.3	80.2±2.4	74.4±3.9	84.0±2.2	78.2±2.3	82.2±2.4	**84.7±2.3**	**87.8±1.6**
κ	61.8±4.7	69.0±5.1	64.7±4.3	69.1±3.6	68.6±5.1	75.6±3.9	70.5±6.2	75.3±5.5	62.6±7.1	75.6±2.6	68.7±3.2	74.4±3.3	**78.8±3.1**	**79.9±2.0**

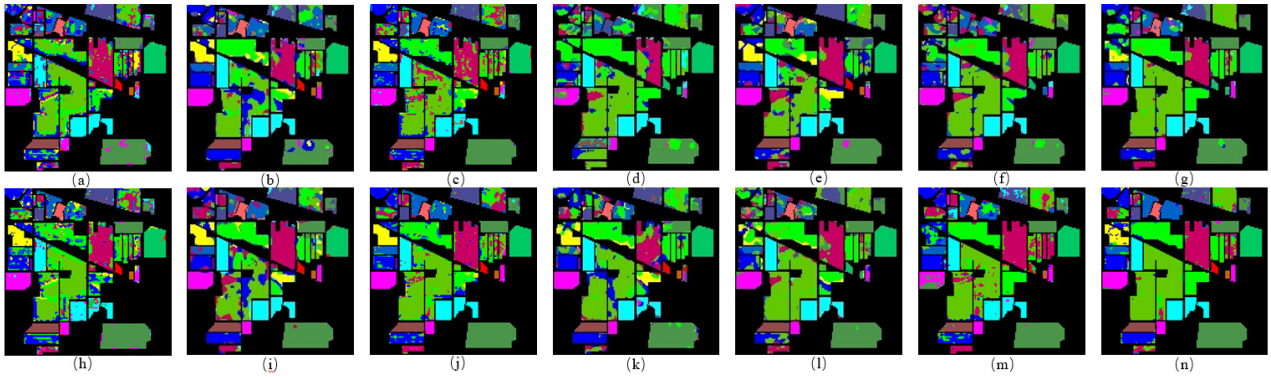


Fig. 9. Classification map of the IP data set. (a) 2D-CNN (L = 5). (b) 3D-CNN (L = 5). (c) DRNN (L = 5). (d) DBMA (L = 5). (e) MSDN (L = 5). (f) DFSL (L = 5). (g) Proposed method (L = 5). (h) 2D-CNN (L = 10). (i) 3D-CNN (L = 10). (j) DRNN (L = 10). (k) DBMA (L = 10). (l) MSDN (L = 10). (m) DFSL (L = 10). (n) Proposed method (L = 10).

TABLE III

CLASSIFICATION RESULTS (%) USING FIVE AND TEN LABELED SAMPLES FOR EACH CLASS ON THE UP DATA SET (THE BOLD VALUE IS THE BEST ACCURACY IN EACH CASE)

Class	2D-CNN		3D-CNN		DRNN		DBMA		MSDN		DFSL		SSAD(Ours)	
	L=5	L=10	L=5	L=10	L=5	L=10	L=5	L=10	L=5	L=10	L=5	L=10	L=5	L=10
1	64.6±6.0	74.9±5.6	65.6±8.5	78.9±3.3	62.4±5.1	83.5±3.4	73.9±5.8	82.0±3.5	65.9±5.9	68.5±7.8	72.8±6.1	84.1±4.5	82.5±2.2	86.6±1.9
2	68.9±4.4	76.2±3.3	71.9±4.1	78.5±4.3	74.2±3.8	78.9±3.1	79.9±9.8	79.5±4.5	75.8±5.3	78.8±5.3	80.3±5.1	80.1±5.6	80.3±2.8	84.5±2.4
3	61.8±5.5	75.4±3.8	63.4±7.2	80.8±3.4	55.4±4.9	86.4±3.2	66.4±8.5	75.7±3.7	63.4±6.5	66.1±7.4	56.3±6.2	76.7±4.8	85.0±2.3	82.2±2.0
4	76.6±4.6	85.0±3.6	81.7±6.5	79.4±3.2	82.3±3.6	81.9±2.0	90.0±5.8	88.2±3.6	86.0±4.9	87.6±6.5	96.7±2.5	89.6±4.6	83.6±2.5	91.6±1.9
5	79.0±3.6	86.8±2.8	83.8±3.1	79.4±4.3	86.2±2.2	81.2±2.7	91.0±5.9	91.2±4.4	91.2±2.0	91.3±2.4	97.9±2.4	90.1±2.6	83.9±2.8	93.9±2.4
6	68.6±5.4	71.1±4.2	67.2±6.8	80.9±3.8	71.4±5.3	73.1±4.8	75.3±6.0	85.0±3.9	81.8±3.2	84.3±6.2	73.8±6.4	85.4±7.0	84.2±2.4	89.4±2.1
7	73.6±4.6	81.3±3.5	81.5±5.6	80.8±3.9	80.1±3.9	78.8±2.6	83.0±5.1	88.4±4.1	88.5±4.5	84.0±3.4	85.8±4.8	89.4±3.7	83.9±2.6	91.8±2.2
8	68.7±6.7	75.9±4.9	71.4±8.2	81.0±4.5	76.0±6.2	79.1±4.5	80.5±5.5	84.4±4.5	86.0±3.7	89.4±4.2	80.1±8.4	86.1±5.2	85.1±2.8	89.0±2.4
9	77.5±3.5	86.3±2.9	81.1±2.7	79.7±3.4	84.4±2.2	81.3±2.1	93.3±3.2	90.9±3.5	89.1±2.8	87.9±2.4	92.4±2.3	93.2±2.2	84.8±2.2	90.2±1.9
OA	69.0±4.9	76.7±4.5	71.5±4.2	79.4±3.7	72.6±4.1	79.7±4.1	79.3±4.1	82.3±4.1	77.1±3.8	79.5±3.1	79.3±3.3	83.3±2.9	**82.3±2.5**	**86.8±2.3**
AA	71.0±4.3	79.2±4.1	74.2±3.8	79.9±3.7	74.7±3.8	80.5±3.9	81.5±4.0	85.0±3.9	80.9±3.4	82.0±2.9	81.8±3.2	86.1±2.8	**83.7±2.6**	**88.8±2.1**
κ	68.4±5.1	74.8±4.7	67.9±4.8	76.2±4.2	71.9±4.3	77.3±3.8	75.1±4.6	79.3±4.4	73.9±4.2	76.0±3.2	75.4±3.7	80.0±3.3	**81.2±2.8**	**83.4±2.4**

knowledge distillation strategy. Similar to the performances in Table V, the results generated by different PC layers are higher than those generated by the "baseline" model. The fused results show better performance than the results generated by the "baseline" method and each single PC layer. These improvements indicate the usefulness of adaptive knowledge distillation.

3) Effectiveness of SSL With Both 3-D Transformation and Adaptive Knowledge Distillation: In this ablation study, the PCN was trained under adaptive knowledge distillation

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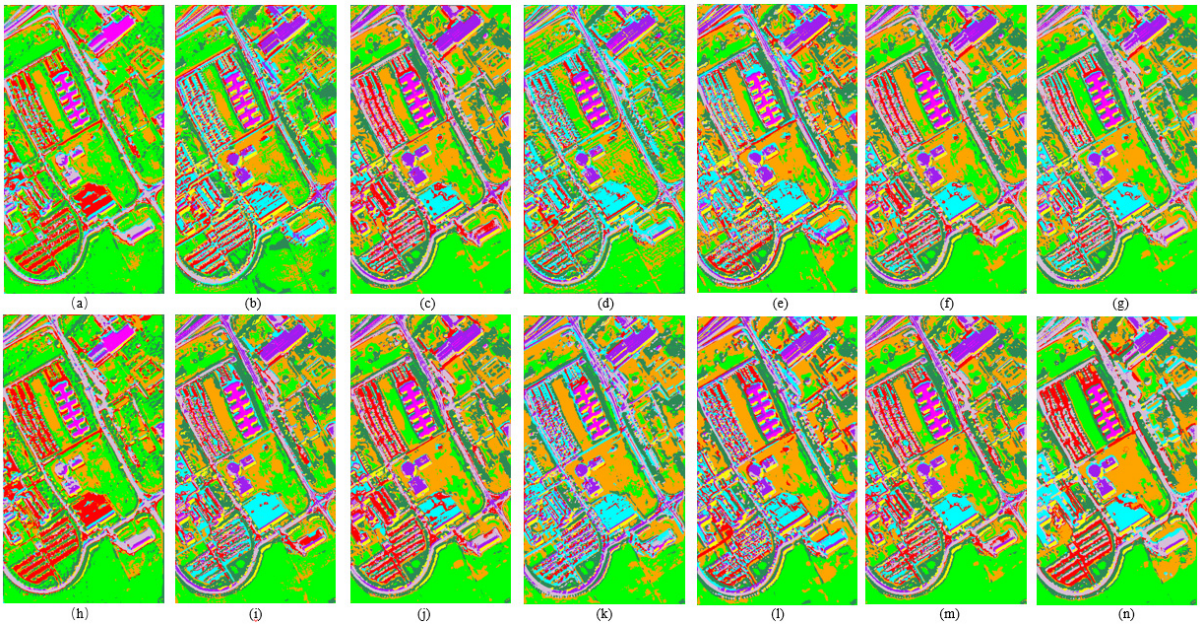


Fig. 10. Classification map of the UP data set. (a) 2D-CNN (L = 5). (b) 3D-CNN (L = 5). (c) DRNN (L = 5). (d) DBMA (L = 5). (e) MSDN (L = 5). (f) DFSL (L = 5). (g) Proposed method (L = 5). (h) 2D-CNN (L = 10). (i) 3D-CNN (L = 10). (j) DRNN (L = 10). (k) DBMA (L = 10). (l) MSDN (L = 10). (m) DFSL (L = 10). (n) Proposed method (L = 10).

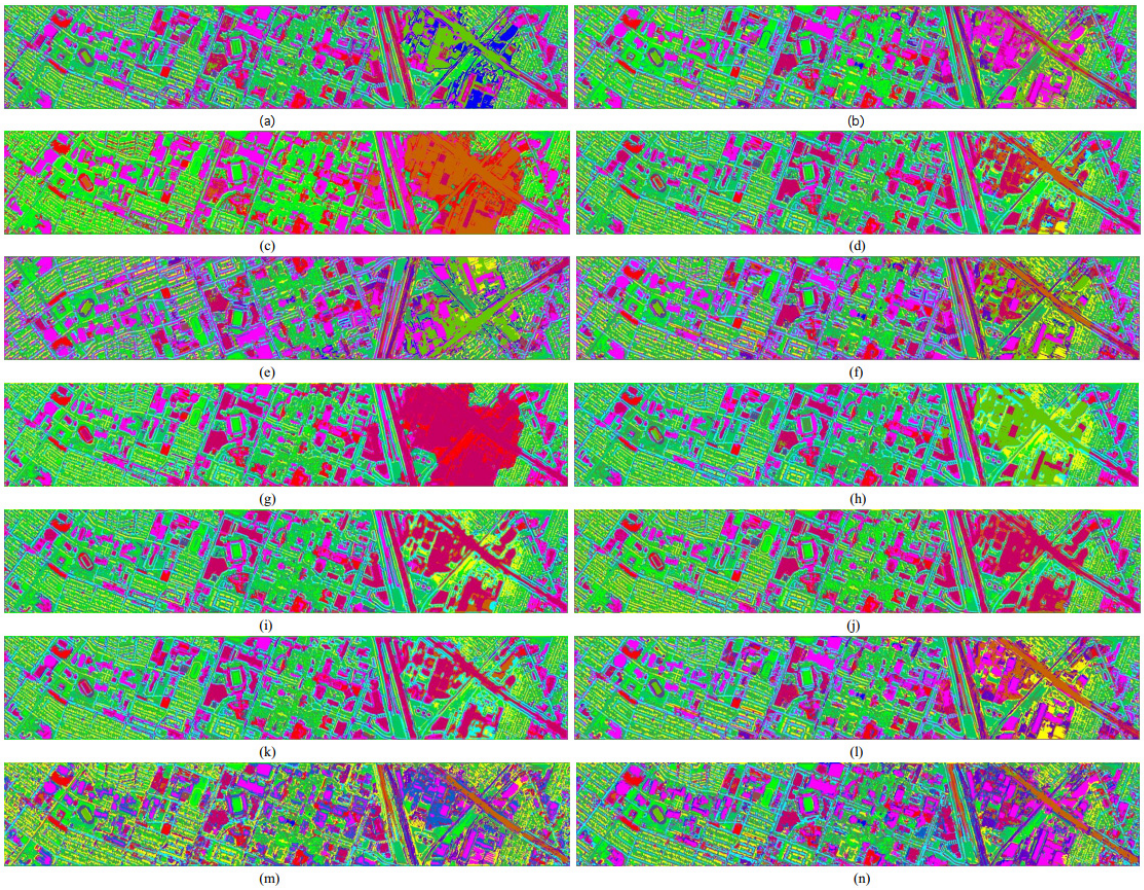


Fig. 11. Classification map of the HS data set. (a) 2D-CNN (L = 50). (b) 2D-CNN (L = 80). (c) 3D-CNN (L = 50). (d) 3D-CNN (L = 80). (e) DRNN (L = 50). (f) DRNN (L = 80). (g) DBMA (L = 50). (h) DBMA (L = 80). (i) MSDN (L = 50). (j) MSDN (L = 80). (k) DFSL (L = 50). (l) DFSL (L = 80). (m) Proposed method (L = 50). (n) Proposed method (L = 80).

with the 3-D transformation strategy. The results of different PC layers ($\{\tilde{\alpha}_k\}_{k=1}^{3}$) and the fused result ($s_o$) over all PC layers are generated by (4) and (5), respectively. The results are shown in Table VII, where "w/ 3DT&AKD" means the PCN is trained under adaptive knowledge distillation with 3-D transformation strategy. Compared with the results

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TABLE IV

CLASSIFICATION RESULTS (%) USING 50 AND 80 LABELED SAMPLES FOR EACH CLASS ON THE HS DATA SET (THE BOLD VALUE IS THE BEST ACCURACY IN EACH CASE)

Class	2D-CNN		3D-CNN		DRNN		DBMA		MSDN		DFSL		SSAD(Ours)	
	L=50	L=80	L=50	L=80	L=50	L=80	L=50	L=80	L=50	L=80	L=50	L=80	L=50	L=80
1	77.1±2.7	80.6±4.6	84.6±3.5	86.8±1.6	83.7±2.5	84.2±4.2	84.7±1.6	90.3±2.8	84.0±1.5	85.5±3.8	87.6±3.0	90.3±1.7	89.0±2.1	86.3±1.4
2	77.8±4.1	79.7±5.0	81.4±4.1	87.4±1.1	78.0±4.6	81.8±3.0	85.6±1.0	87.9±2.6	84.9±1.0	82.4±2.4	86.5±1.9	84.5±1.5	87.8±1.9	89.0±1.9
3	81.5±1.8	83.8±1.5	85.5±1.8	88.8±0.7	82.9±2.0	91.0±2.5	86.6±0.5	95.9±1.1	85.9±1.7	91.3±0.4	91.6±0.4	99.9±0.1	96.9±1.3	92.6±0.7
4	75.3±4.1	72.0±3.3	80.6±4.8	85.4±1.5	74.2±4.2	79.7±4.7	84.5±1.3	85.0±3.2	84.0±1.4	80.9±3.3	84.9±2.5	80.1±1.6	87.6±2.0	90.0±1.9
5	77.7±3.8	84.9±2.9	80.0±2.8	87.2±1.1	80.4±4.0	86.4±2.9	84.7±1.0	88.8±2.0	84.1±1.1	84.0±2.8	86.8±2.0	87.3±1.5	85.9±1.9	91.9±0.7
6	80.9±3.9	84.9±3.5	86.4±3.4	87.4±1.3	82.3±1.7	88.1±2.3	85.7±1.4	93.4±2.3	85.0±1.7	87.4±2.4	89.1±1.9	95.8±1.3	88.4±1.6	90.8±1.2
7	70.0±3.8	71.4±5.1	73.6±4.6	77.3±3.1	77.1±5.3	77.6±5.9	74.5±2.8	77.7±4.0	74.2±3.0	80.1±1.4	79.2±2.4	76.6±2.4	85.3±3.1	87.0±2.0
8	67.0±4.3	76.6±5.6	67.9±4.7	77.5±2.4	66.9±6.5	76.1±4.4	73.5±3.2	76.6±3.8	72.5±3.9	69.0±4.2	73.8±3.5	74.0±2.2	72.2±2.9	84.1±2.0
9	66.2±7.3	80.0±4.2	68.4±4.7	72.8±4.6	72.9±3.3	84.9±5.4	68.4±5.1	72.2±5.7	68.5±5.2	74.3±3.1	72.8±4.8	72.8±4.1	83.7±5.3	84.4±2.6
10	71.5±4.3	76.5±3.7	77.1±3.8	80.8±3.5	81.5±4.2	84.2±3.3	78.2±3.0	85.5±3.9	77.7±3.1	81.5±1.4	82.0±2.7	88.3±2.4	90.2±3.1	89.8±1.8
11	68.5±5.4	71.3±2.9	71.9±4.0	75.4±3.4	73.7±4.0	80.9±4.9	71.5±3.9	79.5±4.4	71.8±4.1	76.7±2.0	75.4±3.3	85.0±2.9	89.0±3.8	87.3±2.3
12	66.5±6.1	68.2±3.7	67.9±4.6	74.0±4.4	70.3±4.2	77.5±4.4	69.9±4.0	74.1±4.6	69.8±4.4	76.3±3.9	74.7±4.3	74.7±3.4	81.5±4.4	89.0±1.6
13	57.8±6.2	88.0±4.1	64.2±5.0	65.9±4.2	81.9±5.9	88.0±4.1	57.7±4.9	68.0±5.5	58.3±5.2	73.6±5.3	66.2±5.6	77.7±3.6	88.4±4.6	85.0±2.8
14	79.9±3.7	85.9±3.3	83.7±2.2	88.6±0.7	83.7±2.8	90.3±2.5	86.6±0.7	95.0±1.5	85.9±1.2	89.0±2.4	90.4±1.6	98.1±1.3	87.1±1.7	92.3±0.7
15	79.1±4.8	84.1±3.4	81.0±2.2	88.8±1.1	81.6±3.5	89.3±3.4	86.5±1.0	92.7±1.8	85.7±1.0	87.9±3.3	89.9±2.3	95.1±1.5	85.1±1.9	92.2±0.8
OA	72.5±4.7	77.5±3.4	76.2±3.7	81.1±3.5	76.9±3.8	82.6±3.5	78.2±2.5	83.0±3.1	77.7±2.6	80.2±3.3	81.3±2.4	83.3±2.9	**85.9±2.2**	**88.3±1.7**
AA	73.1±3.8	79.2±3.2	76.9±2.1	81.6±2.3	78.1±3.5	84.0±2.9	78.6±2.4	84.2±2.5	78.2±2.6	81.3±2.8	82.1±1.7	85.3±3.8	**86.5±1.8**	**88.8±1.6**
κ	71.2±4.1	77.4±4.0	75.0±3.9	80.3±3.9	75.9±3.7	81.4±3.4	77.3±3.0	81.8±3.3	77.0±2.8	79.2±3.3	80.3±2.8	81.6±2.7	**84.9±2.5**	**87.9±1.8**

TABLE V

EFFECTIVENESS OF SSL WITH THE 3-D TRANSFORMATION STRATEGY, WHERE "W/ SSL-3DT" MEANS THE NETWORK WAS TRAINED WITH THE 3-D TRANSFORMATION STRATEGY

Methods	$L=5$	$L=10$	$L=15$
Baseline	53.0 ± 3.1	59.1 ± 2.7	66.5 ± 2.4
$\tilde{\alpha}_1$ w/ SSL-3DT	57.7 ± 2.4	61.1 ± 2.9	67.4 ± 2.5
$\tilde{\alpha}_2$ w/ SSL-3DT	58.5 ± 3.9	64.6 ± 1.8	68.1 ± 1.6
$\tilde{\alpha}_3$ w/ SSL-3DT	60.7 ± 2.9	66.3 ± 2.8	70.1 ± 1.9
α_o w/ SSL-3DT	**69.3** ± 3.0	**74.7** ± 1.8	**78.1** ± 1.6

TABLE VI

EFFECTIVENESS OF SSL WITH THE ADAPTIVE KNOWLEDGE DISTILLATION STRATEGY, WHERE "W/ SSL-AKD" MEANS THE PCN TRAINED UNDER THE STRATEGY OF SSL WITH THE ADAPTIVE KNOWLEDGE DISTILLATION

Methods	$L=5$	$L=10$	$L=15$
Baseline	53.0 ± 3.1	59.1 ± 2.7	66.5 ± 2.4
α_1 w/ SSL-AKD	59.3 ± 3.7	60.3 ± 2.8	63.7 ± 2.3
α_2 w/ SSL-AKD	71.1 ± 3.2	77.2 ± 3.3	78.0 ± 2.5
α_3 w/ SSL-AKD	72.1 ± 3.5	80.3 ± 2.7	82.2 ± 2.7
α_o w/ SSL-AKD	**73.2** ± 3.9	**83.5** ± 2.1	**84.6** ± 2.3

TABLE VII

EFFECTIVENESS OF THE PROPOSED METHOD, WHERE "W/ 3DT&AKD" MEANS THE PCN IS TRAINED UNDER ADAPTIVE KNOWLEDGE DISTILLATION WITH 3-D TRANSFORMATION STRATEGY

Methods	$L=5$	$L=10$	$L=15$
Baseline	53.0 ± 3.1	59.1 ± 2.7	66.5 ± 2.4
$\tilde{\alpha}_1$ w/ 3DT&AKD	67.4 ± 3.4	73.3 ± 2.6	73.4 ± 2.2
$\tilde{\alpha}_2$ w/ 3DT&AKD	74.5 ± 2.5	82.7 ± 2.7	81.7 ± 2.2
$\tilde{\alpha}_3$ w/ 3DT&AKD	76.5 ± 3.0	83.5 ± 1.9	83.9 ± 2.3
α_o w/ 3DT&AKD	**80.3** ± 2.5	**84.3** ± 2.0	**86.4** ± 1.5

TABLE VIII

COMPARISONS OF DIFFERENT COMPONENTS IN THE PROPOSED METHOD

Methods	$L=5$	$L=10$	$L=15$
Baseline	53.0 ± 3.1	59.1 ± 2.7	66.5 ± 2.4
α_o w/ SSL-3DT	69.3 ± 3.0	74.7 ± 1.8	78.1 ± 1.6
α_o w/ SSL-AKD	73.2 ± 3.9	83.5 ± 2.1	84.6 ± 2.3
α_o w/ 3DT&AKD	**80.3** ± 2.5	**84.3** ± 2.0	**86.4** ± 1.5

TABLE IX

TESTING TIME OF THE PROPOSED METHOD AND COMPARISON METHODS

Methods	3D-CNN	DBMA	MSDN	DFSL	Ours
Time(s)	13.67	13.56	12.46	11.13	10.82

generated by "baseline" method, the improvements of our final fusion results for $L = 5$, 10, and 15 are 27.3%, 25.2%, and 19.9%, respectively, showing the effectiveness of the proposed method.

To clearly show the effectiveness of different strategies in our proposed method, the final results of the PCN trained with different strategies are listed in Table VIII. The results show that the combination of these strategies helps the PCN to achieve the best results over all settings. This ablation study illustrates the complementarity between adaptive soft labels and the 3-D transformation strategy.

C. Time Consumption

The time consumption of the proposed method and the comparison methods was tested on the IP data set. The detailed results are shown in Table IX. The computer environment for

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time testing in this article is shown as follows: the processor is "Intel(R) Xeon(R) Gold 5118"; the graphics card is "NVIDIA GeForce RTX 2080 Ti" with "CUDA version 10.0.130"; the programming language and the deep learning platform are Python (version 3.6.2) and PyTorch (version 1.2.0), respectively. Table IX shows that the time consumption for testing the whole IP data set of the proposed method is similar to those of other methods under the same computing environment.

V. Conclusion

This study proposed an SSL method with adaptive distillation, including the strategies of adaptive knowledge distillation and data transformation in spectral and spatial domains, for HSI classification with a small number of labeled samples. The experiments were conducted for verification, and some main conclusions were reached as follows:

1) The proposed method outperforms the existing popular HSI classification methods in the presence of small annotated training samples. The comparative analysis of results suggests that the proposed method achieves state-of-the-art results for the few-shot HSI classification.
2) The ablation study demonstrates that both modules (SSL with adaptive knowledge distillation and SSL with 3-D transformation) are effective for improving the accuracy.
3) The fusion strategies improve the performance of pixel-level classification under the limited labeled training samples.

The SSL method proposed in this article boosts the existing SSL methods by two modules: SSL with 3-D transformation and SSL with adaptive knowledge distillation. Next, we will explore the SSL method based on nonlinear transformation and generative adversarial networks.

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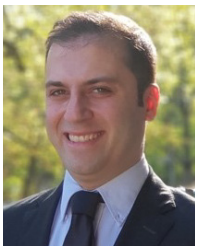
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